RITIK KUMAR

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Summary

Backend-focused full-stack developer (Node.js, TypeScript, PostgreSQL, Redis, AWS) seeking SDE / Backend Developer roles. Shipped an email scheduling engine (3.8x faster DB writes) and a real-time collaborative workspace with optimistic concurrency control; interned at a career-tech company serving a large user base. 300+ DSA problems solved.

Education

Indian Institute of Information Technology, Ranchi 2023 – 2027
B.Tech, Electronics and Communication Engineering *CGPA: 7.39 (till 6th semester)*

Technical Skills

Programming Languages: JavaScript (ES6+), TypeScript, SQL, C++, HTML/CSS
Frontend: React.js, Vite, Tailwind CSS
Backend: Node.js, Express.js, REST APIs, WebSockets (Socket.io), JWT, OAuth 2.0, Zod
Databases & Queues: PostgreSQL, MongoDB (Mongoose), Redis, Prisma ORM, BullMQ
Tools & Platforms: Git, GitHub, GitHub Actions (CI), Docker, AWS (EC2, S3), Vercel, Render, Linux, Postman
Testing & Performance: Vitest, Supertest, node:test, Autocannon (load testing)
Core Fundamentals: Data Structures & Algorithms (300+ Solved), OOP, DBMS, OS, Computer Networks

Experience

Briscent Global LLC – Web Development Intern (resume and career consultancy) Remote | Mar 2026 – Jun 2026
React.js, TypeScript, Node.js, REST APIs

- Developed a resume upload module and a client-facing hiring interface for a resume-rating and hiring platform with a large user base; delivered 15+ full-stack features (React, TypeScript, Node.js REST APIs) working directly with clients.
- Improved Largest Contentful Paint (LCP) by ∼35% and reached 90+ Lighthouse Performance scores on tested pages using route-level code splitting, lazy-loaded imports, and React memoization.
- Engineered reusable API hooks with async state management, client-side caching, and bounded exponential-backoff retries; tested and debugged Node.js REST endpoints with Postman.
- Wrote unit tests for critical UI and API states and enforced ESLint standards across reusable components.

Projects

HyperSend – Email Scheduling Engine hypersend.vercel.app
TypeScript, Node.js, Express, BullMQ, Redis (AOF), PostgreSQL, Prisma, React, Docker github.com/Ritik-7032/Email-Job-Scheduler

- Architected a rate-limited email scheduler (BullMQ, Redis, PostgreSQL) with separate API and worker processes for up to 10,000 recipients; cut DB write time from 8.5 s to 2.2 s (3.8x, remote Postgres) with a single-roundtrip `unnest` insert.
- Designed an atomic Redis Lua rate limiter (2s per-sender gap, hourly quota) that prevents race conditions across workers; throttled jobs reschedule via BullMQ `moveToDelayed` without using retry attempts.
- Prevented duplicate sends with stable job IDs, atomic SQL claims, and a Redis dispatch ledger; reconcilers recover from API, worker, and Redis failures (3-worker `kill -9` test: 40/40 delivered, 0 duplicates).
- Wrote 60 unit and integration tests (Vitest, Supertest; ∼79% coverage, 100% on rate limiter, queue, and ingestion modules) with GitHub Actions CI.

CollabDesk – Real-Time Kanban Workspace collab-desk-nine.vercel.app
React 19, Node.js, Express, Socket.io, MongoDB Atlas, Redis, AWS (EC2, S3), Docker, Gemini AI github.com/Ritik-7032/collabDesk

- Built a real-time Kanban platform (React 19, Express, MongoDB, Socket.io) with JWT-authenticated sockets and fail-closed workspace room authorization; added a Redis adapter for multi-instance event fan-out.
- Prevented lost updates in concurrent card edits with optimistic concurrency control (per-task version counter, atomic `findOneAndUpdate`), returning HTTP 409 Conflict on stale writes.
- Implemented direct-to-S3 presigned uploads (5-min expiry, 10 MB cap, MIME whitelist, SVG blocked) with a `HeadObject` confirm step enforcing workspace key prefixes.
- Raised authenticated read throughput from 34 to 98.6 req/s (2.9x) and cut p50 latency from 181 ms to 69 ms (5-run medians, Autocannon) via a bounded 15s-TTL LRU cache with Redis Pub/Sub invalidation; 24 integration tests in CI.

Achievements

Algorithms: Solved 300+ data structures and algorithms problems on LeetCode (profile linked above).

Academics: Scored above the 95th percentile in JEE Main 2023 among 1 million+ candidates.

Leadership: Independently organized a 2-day Faculty Development Program (50+ professors) and 2 volleyball events.